Multi-Label Land-Cover Recognition Using Sentinel-2 Multispectral Classification

Evaluating the Contribution of Visible, NIR, and SWIR Bands on European Satellite Imagery

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Dataset: BigEarthNet v2.0 (Sentinel-2 Level-2A BOA Surface Reflectance)

Background & Motivation: The Mixed Landscape Challenge

The Single-Label Classification Dilemma

- **Landscapes are inherently mixed:** A single 1.2 km × 1.2 km satellite patch contains pastures bordering woodlots, rivers cutting through fields, and village houses.
- **Single-label forces artificial exclusivity:** Assigning only 'Forest' completely discards secondary classes like pastures or streams.
- **Boundary noise & ambiguity:** Ecotones and mixed pixels create severe labeling conflict in conventional remote sensing.

The Multi-Label Paradigm

- **Independent Bernoulli Formulation:** Each class is treated independently: $y = [y_1, y_2, ..., y_7]$ where $y_k \in \{0, 1\}$.
- **94.4% Empirical Multi-Label Prevalence:** 94.4% of patches in our curated dataset contain at least two land-cover classes.
- **Realistic Ecological Mapping:** Accurately detects co-occurring agricultural parcels, woodlands, and built infrastructure without mutual exclusion.

Research Questions & Project Objectives

Central Research Question

Can multispectral Sentinel-2 observations (Visible, NIR, SWIR) accurately resolve multiple co-occurring surface types, and how does expanding spectral dimensionality affect multi-label classification accuracy on small curated datasets?

Controlled Band Ablation

Systematically compare 3 spectral configurations under identical architectures and splits:

- M1 (Visible RGB): Bands B02, B03, B04 (3 ch)
- M2 (Visible + NIR): Bands B02, B03, B04, B08 (4 ch)
- M3 (Full Multispectral): B02, B03, B04, B08, B11, B12 (6 ch)

Evaluation & Minority Diagnostics

Measure Macro-F1, Micro-F1, and Exact Match subset accuracy. Specifically assess the behavior of minority classes (Wetland, Shrub) and evaluate decision threshold calibration without data leakage.

Sentinel-2 Spectral Bands & BigEarthNet v2.0 Dataset

Diagnostic Spectral Bands Used

- **B02 (Blue - 490 nm, 10m):** Sensitive to atmospheric haze and water clarity.
- **B03 (Green - 560 nm, 10m):** Captures green chlorophyll peak and vegetation vigor.
- **B04 (Red - 665 nm, 10m):** Maximum chlorophyll-a absorption; defines red edge.
- **B08 (NIR - 842 nm, 10m):** Internal spongy mesophyll leaf scattering; total absorption by open water.
- **B11 (SWIR-1 - 1610 nm, 20m):** Foliar water content, soil moisture, and impervious built-up.
- **B12 (SWIR-2 - 2190 nm, 20m):** Mineral background, dry biomass, and bare soil.

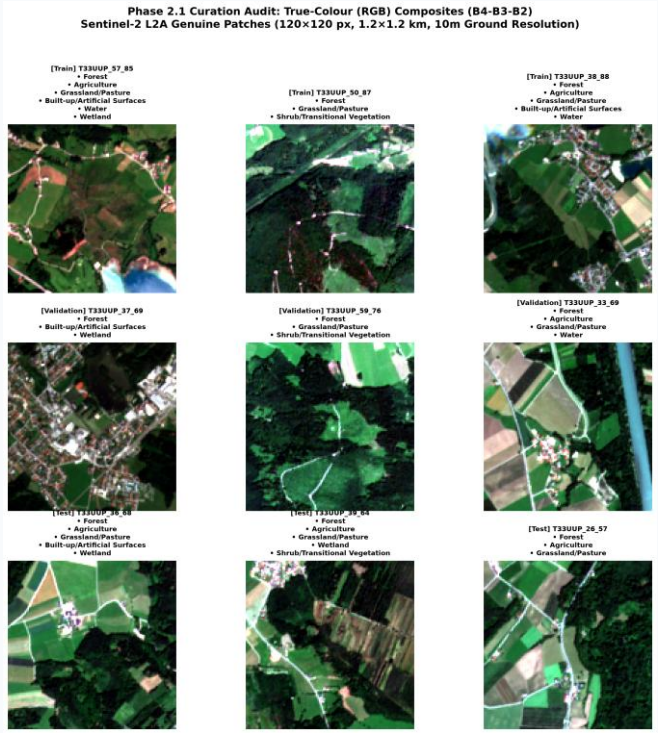
Dataset Geography & Partitions

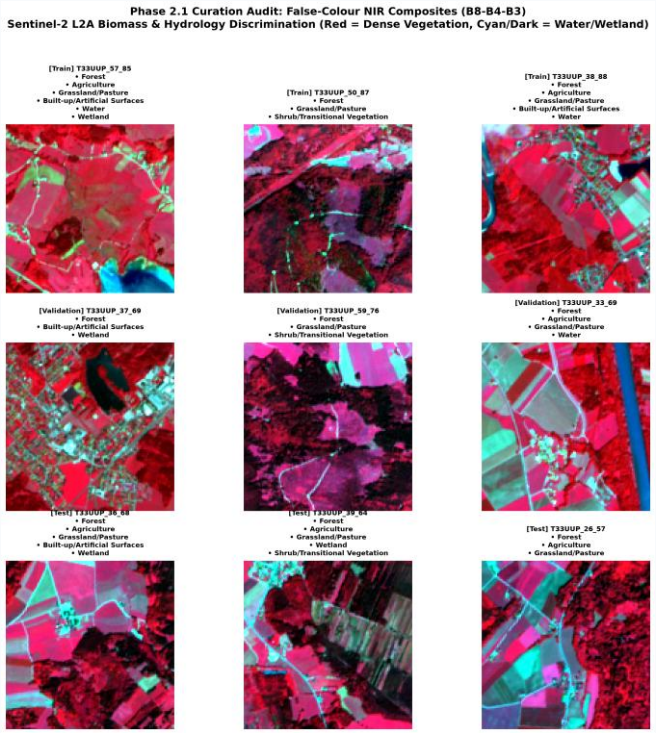
- **Provenance:** BigEarthNet v2.0 (reBEN), Sentinel-2 Level-2A (Sen2Cor atmospheric correction).
- **Geographic Origin:** Tile T33UUP in Austrian Central Alps (UTM Zone 33N).
- **Dataset Size:** 500 genuine patches (1.2 km × 1.2 km, 120×120 pixels).
- **Official Splits:** 300 Train (60%), 100 Validation (20%), 100 Test (20%).
- **Zero Data Leakage:** Strictly disjoint partitions; 0 duplicate patch IDs.
- **Inspection Patches:** All 5 Phase 1 inspection patches preserved strictly in Test.

Reflectance Preprocessing & Multispectral Views

Harmonization Pipeline

- **Bilinear Resampling:** Resampled 20m bands (B11, B12) from 60×60 to 120×120 to align with 10m bands without blocky artifacts.
- **Surface Reflectance Scaling:** ρ = clip(DN / 10000.0, 0.0, 1.0) maps raw DN to physical BOA reflectance [0, 1].
- **Specular Glint Handling:** Rare reflections > 10,000 DN (<0.0042% of pixels) cleanly bounded without distorting dynamic range.
- **True Colour vs False Colour:** CIR (B8-B4-B3) renders dense biomass in vibrant red, while open water absorbs NIR and appears jet black.





Seven Target Classes & Label Distribution

Target Class Name	Train (N=300)	Val (N=100)	Test (N=100)	Dataset Prevalence
Forest	258 (86.0%)	84 (84.0%)	95 (95.0%)	437 (87.4%) - Dominant
Agriculture	142 (47.3%)	74 (74.0%)	52 (52.0%)	268 (53.6%) - Widespread
Grassland / Pasture	209 (69.7%)	75 (75.0%)	39 (39.0%)	323 (64.6%) - Alpine meadows
Built-up / Artificial	130 (43.3%)	58 (58.0%)	31 (31.0%)	219 (43.8%) - Settlements
Water	41 (13.7%)	28 (28.0%)	21 (21.0%)	90 (18.0%) - Lakes & rivers
Wetland	**8 (2.7%) [RARE]**	16 (16.0%)	27 (27.0%)	51 (10.2%) - Severe imbalance
Shrub / Transitional	59 (19.7%)	13 (13.0%)	42 (42.0%)	114 (22.8%) - Clear-cuts/scrub

CNN Architecture & Multi-Label Training Methodology

MultispectralCNN Architecture

- **Input Flexibility:** Dynamically accepts C ∈ {3, 4, 6} channels at 120×120 resolution.
- **4 Convolutional Blocks:** [32, 64, 128, 256 filters] with Batch Normalization, ReLU, and Max Pooling (120→60→30→15→7).
- **Spatial Pooling & Dropout:** AdaptiveAvgPool2d(1, 1) + dual dropout (p=0.4, 0.3) to prevent overfitting on 300 patches.
- **Parameter Count:** Lightweight model (~617k parameters), ideal for small remote sensing datasets.
- **Independent Logits:** Outputs 7 raw unbounded logits, mapped via Sigmoid to [0, 1].

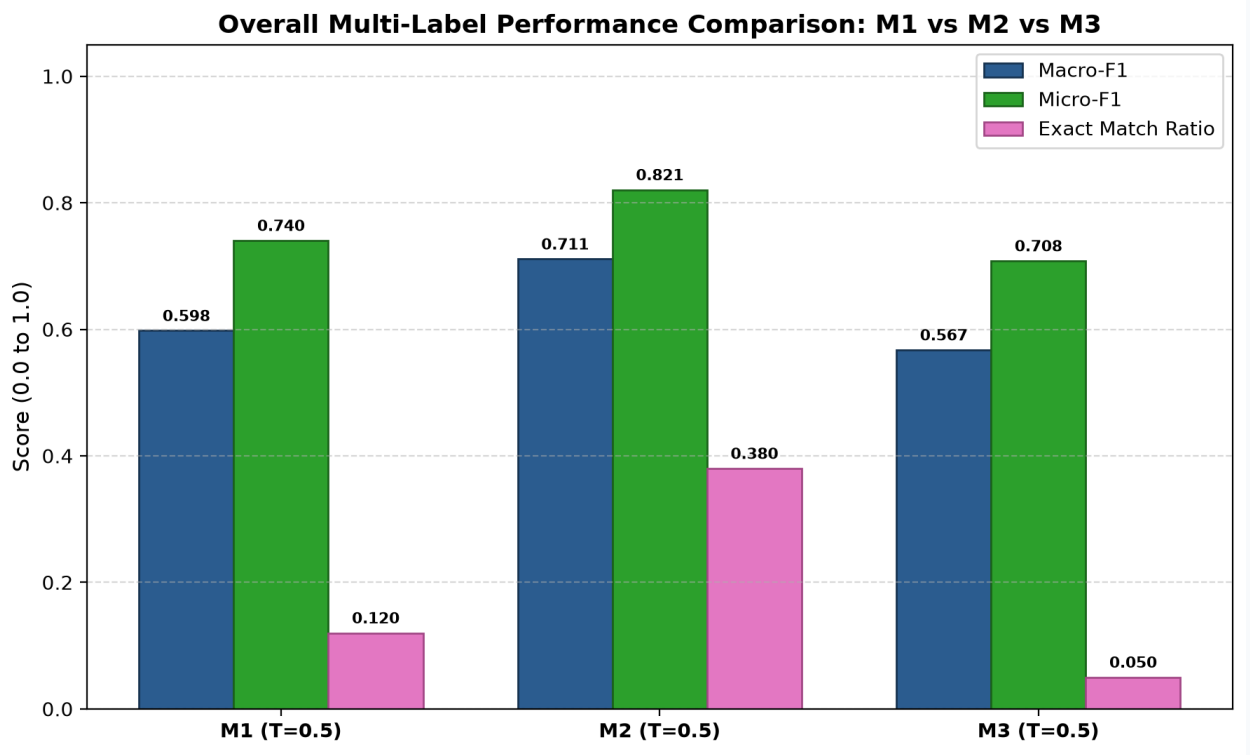
Training Protocol & Validation Calibration

- **Loss Function:** BCEWithLogitsLoss evaluates 7 independent binary cross-entropy losses simultaneously.
- **Optimizer & LR Schedule:** AdamW (lr=1e-3, weight_decay=1e-4) with CosineAnnealingLR (T_max=35).
- **Validation Early Stopping:** Monitored validation loss with patience of 8 epochs (saved best weights).
- **Threshold Calibration:** Tuned per-class thresholds over [0.2, 0.8] strictly on validation data.
- **Zero Test Leakage:** Model weights, normalization, and decision thresholds NEVER touched test labels.

Experimental Results: M1 vs M2 vs M3 Band Ablation

Test Evaluation Results (T=0.5)

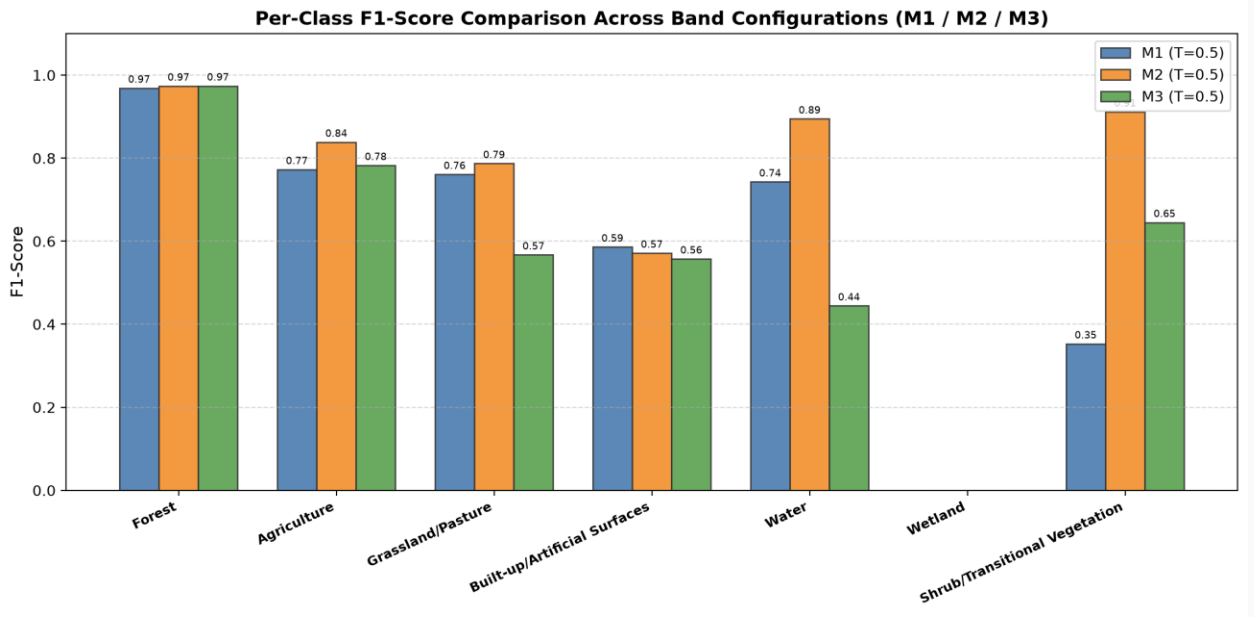
- **M1 (Visible RGB):** Macro-F1 = 0.5978, Micro-F1 = 0.7403, Exact Match = 12.0%. Baseline optical model.
- **M2 (Visible + NIR):** Macro-F1 = 0.7110 (+11.32 pp), Micro-F1 = 0.8205 (+8.02 pp), Exact Match = 38.0% (+26 pp). WINNER!
- **M3 (Multispectral):** Macro-F1 = 0.5673, Micro-F1 = 0.7081, Exact Match = 5.0%. The six-band model (M3) underperformed compared with the RGB+NIR model (M2) on the observed test metrics. Possible explanations include the small training sample, spectral collinearity, resampling effects, and optimization instability. These explanations were not independently tested.
- **Validation-Tuned:** M1 Tuned = 0.7076, M2 Tuned = 0.7020, M3 Tuned = 0.6559. Calibrating thresholds closed M1's gap by lowering cutoffs on minority classes.



Per-Class Performance, Error Analysis & Wetland Failure

Key Per-Class Findings

- **Forest & Agriculture:** Very stable across all models (Forest F1 > 0.96; Agriculture F1 ~0.77-0.84).
- **Shrub Breakthrough in M2:** F1 surged from 0.3529 (M1) to 0.9114 (M2)! NIR mesophyll scattering separates scrub from mature tree canopy.
- **Water Recognition in M2:** F1 rose from 0.7429 (M1) to 0.8947 (M2) with 100% precision due to NIR water absorption.
- **M3 Grassland Over-Prediction:** M3 produced 57 False Positives on Grassland (precision collapsed to 40%), explaining M3's lower score.
- **The Wetland Failure (0.0000 F1):** All 3 models scored 0 TP on Wetland at T=0.5. With only 8 positive training patches (2.7%), unweighted BCE suppressed predictions.



Conclusions, Limitations & Future Directions

Key Scientific Conclusions

• Multi-label recognition is vital: 94.4% of patches in our curated dataset contain at least two land-cover classes.

• NIR is the most valuable spectral addition: M2 (Visible+NIR) gained +11.32 pp Macro-F1 and tripled Exact Match (38%).

• Performance of M3: The six-band model (M3) performed worse than the RGB+NIR model (M2) on the observed test metrics. Small sample size, spectral collinearity, resampling effects, and optimization instability are possible explanations, but they were not independently tested.

Project Limitations

• Single-Tile Scope: All 500 patches originate from Tile T33UUP in the Austrian Alps. Findings do not generalize globally.

• Severe Minority Imbalance: Wetland had only 8 training examples, causing zero-recall.

• 20m Spatial Interpolation: SWIR bands were bilinearly resampled, lacking native 10m high-frequency detail.

Future Research Directions

• Positive-Class Weighting / Focal Loss: Compensate for minority classes like Wetland.

• Transfer Learning: Pre-train on the full continental BigEarthNet archive before fine-tuning.

• Geographic Expansion: Evaluate generalization across Mediterranean, Scandinavian, and arid biomes.